



Faculty of Engineering & Technology Electrical & Computer

Engineering Department

ENCS3340

Project1

Machine Learning Classification Project

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Introduction

Machine Learning is a field of artificial intelligence that enables systems to learn from data and make predictions. In this project, two classification algorithms, Decision Tree and Naive Bayes, are implemented and evaluated using Python and the Scikit-learn library. The main objective is to compare the performance of these models on the PlayTennis dataset and analyze their behavior using different evaluation metrics such as accuracy, precision, recall, and F1-score. In addition, the models are further evaluated using the Iris dataset from the UCI Machine Learning Repository to demonstrate classification performance on a larger and more complex dataset.

Part A: Decision Tree

Dataset Description

Table 1: PlayTennis Dataset Summary

Dataset	PlayTennis
Instances	14
Features	Outlook, Temp, Humidity, Wind
Target Classes	Yes, No
Evaluation Method	Leave-One-Out Cross Validation (LOOCV)

Methodology

The Decision Tree classifier was implemented using the Scikit-learn library. Instead of using a traditional train-test split, Leave-One-Out Cross Validation (LOOCV) was applied due to the small dataset size (14 instances). In LOOCV, the model is trained on 13 samples and tested on 1 sample, and this process is repeated 14 times so that each instance is used once as a test sample. Categorical features were encoded into numerical values before evaluation. The final performance was obtained by aggregating predictions from all LOOCV folds. The model was evaluated using accuracy, confusion matrix, precision, recall, and F1-score. Decision Tree visualization was generated using the full dataset for interpretability.

Decision Tree Visualization

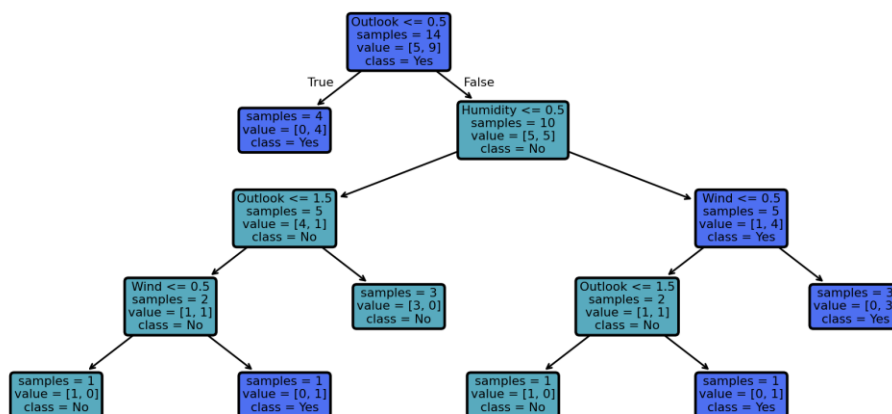


Figure 1: Decision Tree Structure for PlayTennis Dataset

Results

The Decision Tree classifier achieved an accuracy of **57.1%**

Table 2:Confusion Matrix for Decision Tree Predictions

	Pred Yes	Pred No
Actual Yes	6	3
Actual No	3	2

Table 3:Precision, Recall, and F1-Score for Decision Tree

Class	Precision	Recall	F1-Score
Yes	0.67	0.67	0.67
No	0.40	0.40	0.40

Part B: Naive Bayes

Dataset Description

Table 4:PlayTennis Dataset Summary

Dataset	PlayTennis
Instances	14
Features	Outlook, Temp, Humidity, Wind
Target Classes	Yes, No
Evaluation Method	Leave-One-Out Cross Validation (LOOCV)

Methodology

The Naive Bayes classifier was implemented using the Scikit-learn library. Specifically, the Categorical Naive Bayes model was used to handle categorical features in the PlayTennis dataset. Instead of using a traditional train-test split, Leave-One-Out Cross Validation (LOOCV) was applied due to the small dataset size (14 instances). In LOOCV, the model is trained on 13 samples and tested on 1 sample, and this process is repeated 14 times so that each instance is used once as a test sample. The classifier estimates prior probabilities for each class (Yes/No) and conditional probabilities for each feature given the class. For each test sample, the model computes posterior probabilities using the Naive Bayes assumption of feature independence, and selects the class with the highest probability as the final prediction. The final performance was obtained by aggregating predictions from all LOOCV folds. The model was evaluated using accuracy, confusion matrix, precision, recall, and F1-score.

Results

The Naive Bayes classifier achieved an accuracy of **50%**

Table 5:Confusion Matrix for Naïve Bayes Predictions

	Pred Yes	Pred No
Actual Yes	6	3
Actual No	4	1

Table 6:Precision, Recall, and F1-Score for Naive Bayes

Class	Precision	Recall	F1-Score
Yes	0.60	0.67	0.63
No	0.25	0.20	0.22

Comparison

The bar chart in Figure 2 illustrates the performance comparison between the Decision Tree and Naive Bayes classifiers across four evaluation metrics: accuracy, average precision, average recall, and average F1-score. The results are based on Leave-One-Out Cross Validation (LOOCV), where each instance in the dataset is used once as a test sample.

The figure and Table 7 show that the Decision Tree achieved higher values than Naive Bayes in all evaluation metrics. The Decision Tree achieved an accuracy of **57.1%**, while Naive Bayes achieved **50.0%**.

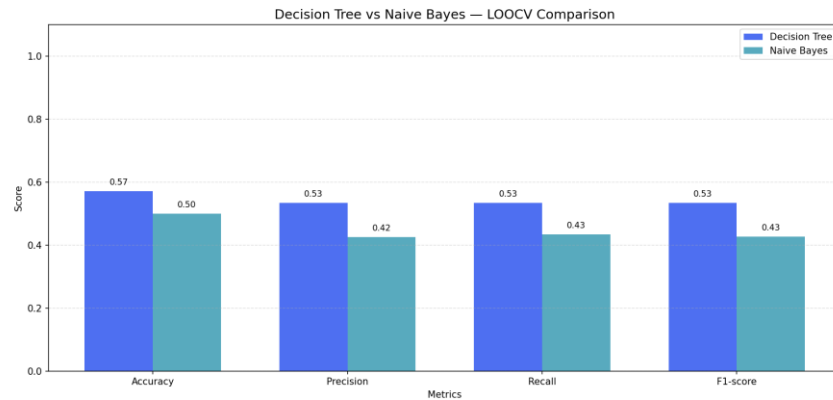


Figure 2:Decision Tree vs Naive Bayes Performance Comparison

Table 7:Performance Comparison Between Decision Tree and Naive Bayes

Metric	Decision Tree	Naive Bayes	Winner
Accuracy	0.571	0.500	Decision Tree
Avg Precision	0.533	0.425	Decision Tree
Avg Recall	0.533	0.433	Decision Tree
Avg F1-Score	0.533	0.427	Decision Tree

The comparison results show that the Decision Tree classifier outperformed the Naive Bayes classifier on the PlayTennis dataset across all evaluation metrics under LOOCV evaluation. This

performance difference can be explained by the ability of Decision Trees to capture feature interactions and decision rules directly. In contrast, Naive Bayes assumes conditional independence between features, which may not hold in this dataset. For example, attributes such as Outlook and Humidity can jointly influence the final decision, which is better captured by the Decision Tree model. Another advantage of the Decision Tree is its interpretability, as it provides a clear tree structure representing decision paths. Naive Bayes, while computationally efficient, is less interpretable and relies on probabilistic assumptions. Overall, the Decision Tree is more suitable for this dataset, as it achieves better performance and provides more meaningful decision rules under LOOCV evaluation.

Part C: Iris Dataset (UCI)

Dataset Description

The [Iris dataset](#) was selected from the UCI Machine Learning Repository. It is one of the most widely used benchmark datasets for classification tasks in machine learning. The dataset contains 150 instances and four continuous numerical features: sepal length, sepal width, petal length, and petal width (all measured in centimeters). The target variable represents the species of iris flowers, which includes three balanced classes: Setosa, Versicolor, and Virginica, with 50 instances per class.

Table 8: Iris Dataset Summary

Dataset	Iris (UCI Machine Learning Repository)
Instances	150 (50 per class)
Features	sepal length, sepal width, petal length, petal width (all in cm)
Target Classes	setosa, versicolor, virginica
Training Size	120 (80%)
Testing Size	30 (20%)

Methodology

In this part, a Decision Tree classifier was applied to the Iris dataset. The dataset was split into training and testing sets using an 80/20 ratio. A Decision Tree model with entropy criterion was trained on the training data. The model learns decision rules by selecting features that provide the highest information gain at each split. After training, predictions were made on the test set and evaluated using accuracy, confusion matrix, precision, recall, and F1-score. Finally, the model performance was analyzed based on classification result

Decision Tree Visualization

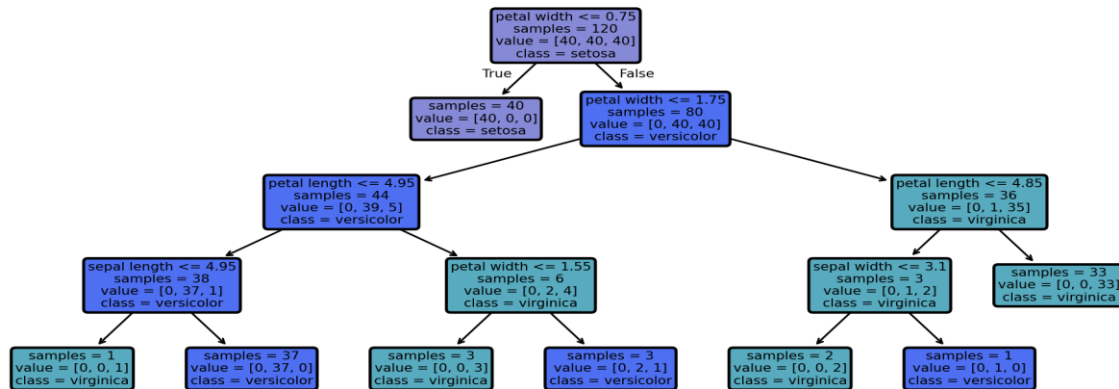


Figure 3: Decision Tree for Iris Dataset

Results

The Decision Tree classifier achieved a high performance on the Iris dataset with an accuracy of **0.967 (96.7%)**, indicating that the model correctly classified most of the test samples.

Table 9: Iris Decision Tree Confusion Matrix

	Pred Setosa	Pred Versicolor	Pred Virginica
Actual Setosa	10	0	0
Actual Versicolor	0	9	1
Actual Virginica	0	0	10

Table 10: Iris Classification Report

Class	Precision	Recall	F1-Score
Setosa	1.00	1.00	1.00
Versicolor	1.00	0.90	0.95
Virginica	0.91	1.00	0.95

Model Performance Discussion

The Decision Tree classifier demonstrated strong overall performance on the Iris dataset. The model achieved high accuracy, indicating that it was able to correctly classify most of the test instances. The balanced precision, recall, and F1-score across classes further confirm that the model performs reliably and consistently.

In terms of strengths, the Decision Tree model is simple, interpretable, and easy to visualize, which makes it highly suitable for understanding classification decisions. It can handle both linear and non-linear relationships between features without requiring feature scaling or normalization. Additionally, it performs particularly well on structured datasets such as Iris, where class boundaries are relatively clear.

However, the model also has some limitations. One of the main weaknesses is its tendency to overfit the training data if the tree becomes too deep or overly complex. Another limitation is that Decision Trees can be sensitive to small variations in the dataset, which may lead to changes in the tree structure and affect stability.

From the observed results, the model performed exceptionally well on the Iris dataset, especially in correctly classifying the Setosa class. Minor misclassifications occurred between Versicolor and Virginica due to their similarity in feature space. Overall, the results show that the Decision Tree classifier is highly effective for this dataset and benefits from the clear separation between classes.

Conclusion

In this project, two classification algorithms, Decision Tree and Naive Bayes, were implemented and evaluated using the PlayTennis dataset under Leave-One-Out Cross Validation (LOOCV), followed by an additional evaluation of the Decision Tree model on the Iris dataset. The main objective was to compare the performance of different machine learning models and understand their behavior across different types of data.

The results showed that the Decision Tree classifier consistently outperformed the Naive Bayes classifier on the PlayTennis dataset, achieving higher accuracy and better overall performance across all evaluation metrics. This is mainly due to its ability to capture relationships and interactions between features, whereas Naive Bayes assumes feature independence, which may not always hold in real-world data.

Furthermore, the Decision Tree model achieved very high accuracy on the Iris dataset (96.7%), demonstrating its effectiveness when applied to larger, continuous, and well-structured datasets. This clearly indicates that model performance is strongly influenced by dataset characteristics such as size, feature type, and data distribution.

The comparison between Decision Tree and Naive Bayes shows that Decision Tree is more effective for this problem, as it can model feature interactions directly. In contrast, Naive Bayes relies on a strong independence assumption, which can reduce its performance when features are correlated. Additionally, Decision Tree models are more interpretable due to their tree structure, while Naive Bayes is simpler but less intuitive in explaining decision-making.

Despite their advantages, both algorithms have certain limitations. Decision Trees are prone to overfitting, especially when trained on small datasets such as PlayTennis, where small changes in the data can significantly affect the structure of the model. On the other hand, Naive Bayes may underperform when the independence assumption between features is violated. In general, both models are highly dependent on dataset quality and size, and their performance improves significantly when applied to larger and more structured datasets, as demonstrated by the Iris results.

